

TANEGASHIMA, Japan

WMO: 478370

Lat: 30.7204N

Lon: 130.9830E

Elev: 38

StdP: 100.87

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	6.7	-2.5	3.1	8.3	-1.2	3.4	8.7	17.7	9.9	16.3	10.3	7.0	330	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.8	31.8	26.7	31.2	26.4	30.6	26.2	27.6	30.5	27.1	30.0	26.7	29.5	4.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.4	29.5	26.3	21.8	29.1	25.9	21.4	28.9	88.0	30.7	85.9	30.1	84.2	29.6	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.2	13.3	11.7	3.3	33.0	1.3	0.7	2.4	33.4	1.6	33.8	0.9	34.2	0.0	34.6		
(5)			0.7	28.0	1.1	0.6	-0.1	28.4	-0.7	28.8	-1.3	29.1	-2.0	29.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	20.0	11.6	12.3	14.5	18.0	21.4	24.1	27.8	28.4	26.4	22.7	18.2	13.7
	DBStd	6.26	2.79	3.05	2.76	2.31	1.74	2.07	1.42	1.09	1.76	2.24	2.78	3.03
	HDD10.0	30	14	9	2	0	0	0	0	0	0	0	0	4
	HDD18.3	716	209	169	121	34	1	0	0	0	0	1	36	145
	CDD10.0	3669	64	74	142	239	352	422	552	571	493	393	246	121
	CDD18.3	1313	0	1	3	22	95	172	294	313	243	136	32	3
	CDH23.3	10819	0	0	1	19	206	962	3210	3638	2200	545	38	1
	CDH26.7	3023	0	0	0	0	6	140	1026	1301	509	41	1	0
Wind	WSAvg	5.1	6.6	6.1	6.0	5.2	4.5	4.4	4.4	4.3	4.2	4.5	4.8	6.2
Precipitation	PrecAvg	3149	160	174	229	271	299	639	272	204	332	218	197	154
	PrecMax	4076	387	445	430	495	515	1058	718	507	1034	672	368	343
	PrecMin	1532	2	65	85	111	110	265	0	18	57	35	63	30
	PrecStd	505	72	85	84	111	128	199	196	126	221	155	93	69
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.6	21.6	22.5	25.1	27.4	30.5	32.4	33.1	31.8	29.1	26.2	22.2
		MCWB	17.1	18.0	18.1	20.8	22.7	26.3	26.9	27.4	26.7	24.7	22.3	18.4
	2%	DB	18.8	19.9	21.1	23.8	26.2	29.5	31.6	32.1	30.7	27.8	24.3	20.6
		MCWB	15.8	17.3	17.1	19.6	21.6	25.8	26.7	26.8	25.7	23.7	20.6	17.3
	5%	DB	17.4	18.6	20.0	22.7	25.2	28.3	31.0	31.4	30.0	26.8	23.3	19.6
		MCWB	14.3	15.9	16.7	18.9	21.0	25.2	26.5	26.5	25.3	23.0	19.6	16.3
	10%	DB	16.1	17.2	18.9	21.8	24.5	27.2	30.4	30.8	29.2	25.9	22.3	18.4
		MCWB	12.8	14.3	15.6	18.1	20.5	24.5	26.3	26.3	24.9	22.2	18.7	15.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.1	19.2	19.8	22.4	24.2	26.7	28.0	28.3	27.5	25.8	23.2	19.8
		MCDB	19.8	20.8	21.4	24.0	26.0	29.5	31.0	31.3	30.4	27.8	25.4	21.0
	2%	WB	16.4	17.7	18.6	21.2	23.4	26.0	27.5	27.7	26.7	24.9	21.7	18.1
		MCDB	18.2	19.5	20.2	22.7	25.1	28.5	30.3	30.8	29.4	27.0	23.6	20.1
	5%	WB	14.8	16.3	17.4	20.2	22.6	25.5	27.1	27.2	26.2	24.1	20.5	16.8
		MCDB	17.1	18.3	19.3	21.8	24.3	27.7	29.9	30.2	28.8	26.1	22.5	19.1
	10%	WB	13.1	14.8	16.2	18.9	21.7	25.0	26.7	26.9	25.7	23.1	19.5	15.4
		MCDB	15.7	16.9	18.3	21.0	23.4	27.0	29.4	29.7	28.2	25.3	21.7	18.0
Mean Daily Temperature Range	5% DB	MDBR	4.8	5.2	5.5	5.6	5.4	4.4	4.7	4.8	4.8	4.7	5.1	5.1
		MCDBR	6.8	6.7	6.7	7.0	6.3	5.0	5.4	5.6	5.4	5.2	5.8	6.1
		MCWBR	6.1	6.0	5.2	4.6	3.8	2.7	2.4	2.4	2.5	3.1	4.1	5.0
	5% WB	MCDBR	6.1	6.1	5.6	5.1	5.0	4.6	4.9	4.9	4.9	4.3	4.8	5.1
		MCWBR	6.1	6.3	5.5	4.6	4.1	2.8	2.5	2.4	2.6	3.1	4.3	5.2
Clear-Sky Solar Irradiance	taub	0.435	0.469	0.523	0.537	0.534	0.529	0.506	0.497	0.489	0.476	0.440	0.427	
	taud	2.116	2.052	1.918	1.940	2.020	2.126	2.221	2.246	2.226	2.174	2.197	2.139	
	Ebn at Noon	768	780	766	774	777	777	794	797	789	767	760	752	
	Edh at Noon	131	153	186	189	176	157	143	138	136	134	120	122	
All-Sky Solar Radiation	RadAvg	2.29	2.90	3.88	4.62	4.82	4.13	5.75	5.68	4.59	3.61	2.64	2.21	
	RadStd	0.22	0.33	0.28	0.44	0.53	0.54	0.57	0.35	0.48	0.39	0.29	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-0.49	N/A	N/A	N/A	N/A	+8	+62	N/A	N/A
(47) Regional (11 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page